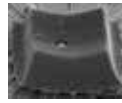




### CRISPR treats genetic disease

CRISPR injected into the blood treats a genetic disease for first time.  
<https://digit.in/jul21-01>



### New material tougher than kevlar

The new carbon-based material could be a basis for tougher, lighter alternatives to steel and kevlar. <https://digit.in/jul21-02>

# Beyond the veil

## Scrying into a time when light didn't travel

Aditya Madanapalle | [aditya@digit.in](mailto:aditya@digit.in)

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his is a uniform glow from space that does not originate from stars. The cosmic microwave

background is the oldest electromagnetic radiation in the universe, from 14 billion years ago. The photons from this period were generated about 400,000 years after the Big Bang, which makes it difficult to learn about the universe before the CMB. During this period,

all photons were trapped in a hot plasma soup, and light did not travel. It was only when it cooled down sufficiently, and the protons and electrons combined to form the first hydrogen atoms, that light started travelling. The CMB is the first visible light of the universe, and in theory, it should be uniform all around, but it isn't. There are some patches in the CMB, remnants of an epoch that is impossible to peer into, and the motley pattern is more prominent on one side of the sky than the other, in other words, the CMB is asymmetrical.

This asymmetry is a clue into the process that happened at the dawn of the cosmos, and are hints of structures that extend into the

unseen universe, clusters and filaments so distant that their light will never reach Earth. As the stars and galaxies recede from our vision with the expansion of the universe, the CMB remains the same, because it permeated all of creation at the very beginning. Furthermore, the asymmetry is just one pattern identified in the CMB, the so called axis of evil, another pattern that may have been caused by exotic physics during the period of inflation.

In 1981 Alan Guth proposed the theory of inflation, which was a theoretical prediction of what happened to the universe immediately after the Big Bang. According to the theory, the universe went through a process known as "inflation",